

MTH212M PRACTICE PROBLEMS WEEK 5

Question 1. Work out the exercises left out in class.

Question 2. Consider a Markov chain $\{X_n\}_{n \geq 0}$ on the state space $\{0, 1, 2, 3\}$ and the T.P. matrix

$$\begin{pmatrix} \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\ 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \end{pmatrix}.$$

For any transient state i and any (closed) recurrent class C , find $\mathbb{P}(X_n \in C \text{ for large } n | X_0 = i)$.

Question 3. Consider a Markov chain $\{X_n\}_{n=0}^\infty$ with state space S and the transition probability matrix P . Find $\lim_{n \rightarrow \infty} p_{ij}^n, \forall i, j \in S$.

(i) $S = \{1, 2, 3, 4\}$ and $P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$

(ii) $S = \{1, 2, 3, 4, 5\}$ and $P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$

Question 4. Prove or disprove: Absorbing states are always positive recurrent.

Question 5. Prove or disprove: In a Markov Chain with finite state space S , the limit $\lim_{n \rightarrow \infty} p_{ii}^n$ exists for all $i \in S$.

Question 6. Consider an aperiodic recurrent irreducible Markov Chain with finite state space. Show that all the states are positive recurrent.

Question 7. In an aperiodic Markov Chain with finite state space, show that at least one state is positive recurrent.